Glance ML Internship Assignment Submission Report

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1. Problem Statement

The goal of this assignment is to build an intelligent search engine that can retrieve specific images from a diverse database based on natural language descriptions. The system needs to understand not just "what" someone is wearing, but "where" they are and the "vibe" of their attire.

2. Approaches Considered & Trade-offs

Approach	**Strengths**	**Weaknesses**
Vanilla CLIP + FAISS	Simple and easy to implement.	Cannot understand compositional queries (e.g., *"red shirt with blue pants"*).
	Strong zero-shot image-text retrieval baseline.	Treats images as a single semantic embedding ("bag of concepts").
	Fast retrieval using a single FAISS index.	Cannot associate attributes (colors) with specific garments.
	Low computational complexity during inference.	Lacks explicit attribute-level reasoning, leading to incorrect retrievals.

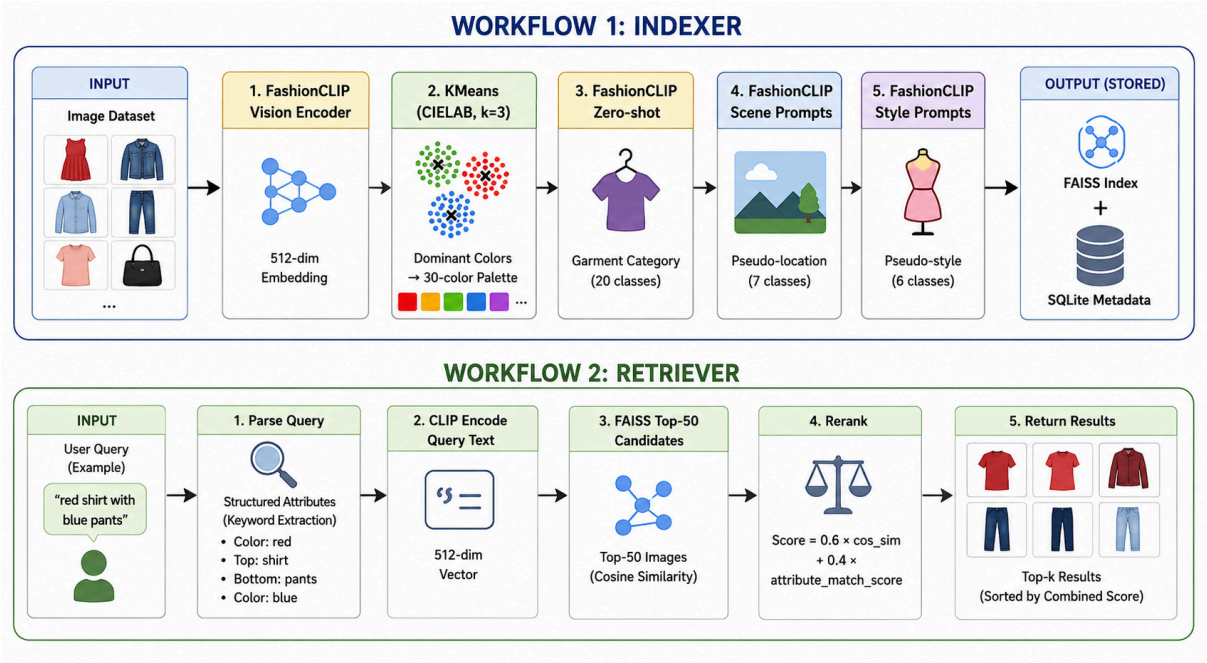
Approach	Strengths	Weaknesses
FashionCLIP + Structured Attribute Reranking (Chosen)	Significantly improves compositional query retrieval by combining semantic similarity with structured attributes.	Pseudo-labels may contain classification errors.
	Explicitly models garment category, color, location, and style.	Rule-based query parsing may not generalize to complex natural language.
	Modular architecture allows independent improvement of each component.	Does not completely solve the garment-color binding problem without object detection.
	Scalable retrieve-then-rerank pipeline suitable for large datasets.	Retrieval quality depends on the accuracy of the generated metadata.

3. Chosen Architecture

System Overview

The proposed system follows a two-stage retrieve-and-rerank architecture consisting of two independent pipelines:

- **Indexing Pipeline**: Processes every image once to extract semantic embeddings and structured metadata, then stores them in a searchable index.
- **Retrieval Pipeline**: Processes the user's query, retrieves the most relevant candidate images using semantic similarity, and reranks them using structured attribute matching.



Indexing Pipeline

During indexing, each image is processed through multiple models to generate both semantic and structured representations.

Component	Purpose
FashionCLIP	Generates a 512-dimensional image embedding for semantic retrieval.
CIELAB KMeans	Extracts the three dominant colors from the image.
FashionCLIP Zero-shot	Predicts the garment category.
Places365 (ResNet50)	Classifies the scene into fashion-relevant location categories.
FashionCLIP Style Classifier	Predicts the clothing style (Formal, Casual, Sporty, etc.).
FAISS + SQLite	Stores image embeddings in FAISS and structured metadata in SQLite for efficient retrieval.

Retrieval Pipeline

When a user submits a query, the system performs retrieval in five sequential steps.

Steps	Description
1. Query Parsing	Extracts structured attributes such as colors, garments, location, and style from the text query.
2. Query Encoding	Converts the query into a 512-dimensional embedding using the FashionCLIP text encoder.
3. Candidate Retrieval	Searches the FAISS vector index and retrieves the Top-50 most semantically similar images.
4. Attribute Reranking	Combines cosine similarity with structured attribute matching to improve retrieval accuracy.
5. Final Results	Returns the highest-ranked images to the user.

4. Dataset

The project uses the **Fashionpedia** dataset, containing **36,498 fashion product images**. The images are catalog-style photographs featuring one or more garments captured under controlled lighting conditions. Since the dataset does not provide annotations for garment category, color, style, or location, all required attributes were generated automatically using pseudo-labeling techniques.

Property	**Description**
Original Dataset	36,498 fashion images
Working Dataset	2,500 images
Selection Method	Stratified random sampling based on garment category and dominant color
Reason for Sampling	Faster indexing and experimentation while maintaining representative category and color distributions

5. Design Decisions

The system was designed by evaluating multiple alternatives for each major component and selecting the approach that provided the best balance between retrieval accuracy, scalability, and implementation complexity

Component	Chosen Approach	Reason for Selection
Image Embedding & Garment Classification	FashionCLIP	FashionCLIP is specifically fine-tuned for fashion images, enabling better recognition of garment categories and fashion terminology than the generic CLIP model. This improves both semantic retrieval and pseudo-label generation.
Color Extraction	CIELAB + K-Means Clustering	CIELAB provides perceptually uniform color representation, while K-Means identifies the dominant garment colors directly from image pixels. This approach produces more reliable color labels than RGB clustering or CLIP-based color prediction.
Location Classification	Places365 (ResNet50)	Places365 is a dedicated scene recognition model capable of accurately classifying indoor and outdoor environments. It provides more reliable location labels than zero-shot vision-language models.
Query Parsing	Rule-Based Keyword Parser	A lightweight keyword-based parser extracts garment, color, location, and style attributes from user queries. This approach is deterministic, fast, and sufficient for the assignment without requiring external language models.

Retrieval Strategy	Retrieve-and-Rerank	The system first retrieves candidate images using semantic similarity (FAISS + FashionCLIP) and then reranks them using structured attribute matching, improving compositional query retrieval while maintaining scalability.
Scoring Function	0.6 × Cosine Similarity + 0.4 × Attribute Match	A weighted combination of semantic similarity and attribute matching provided the best balance between visual relevance and structured attribute reasoning during experimentation.

7. Evaluation

Evaluation Methodology

The proposed retrieval system was evaluated using the five test queries provided in the assignment. Each query was designed to assess a different aspect of multimodal fashion retrieval, including garment recognition, color matching, style understanding, location awareness, and compositional reasoning.

The retrieved results were manually inspected to determine whether the returned images satisfied the requested attributes. This qualitative evaluation provides a clear indication of how well the system performs on real-world fashion search scenarios.

Query	**Expected**	**System Performance**	**Observation**
A person in a bright yellow raincoat	Yellow raincoat	Not Retrieved	The sampled dataset contained very few yellow raincoat images, limiting retrieval performance.
Professional business	Formal outfit in an office environment	Successfully Retrieved	The system correctly identified both the formal

attire inside a modern office			clothing style and indoor office setting.
Someone wearing a blue shirt sitting on a park bench	Blue shirt in a park	Partially Retrieved	Relevant park scenes were found, but the required blue shirt combination was not present in the indexed metadata.
Casual weekend outfit for a city walk	Casual outfit in an urban environment	Successfully Retrieved	The retrieve-and-rerank pipeline accurately matched both clothing style and location.
A red tie and a white shirt in a formal setting	Formal attire with correct garment-color combination	Successfully Retrieved	The attribute reranker significantly improved retrieval by correctly matching garment and color attributes.

8. Failure Analysis

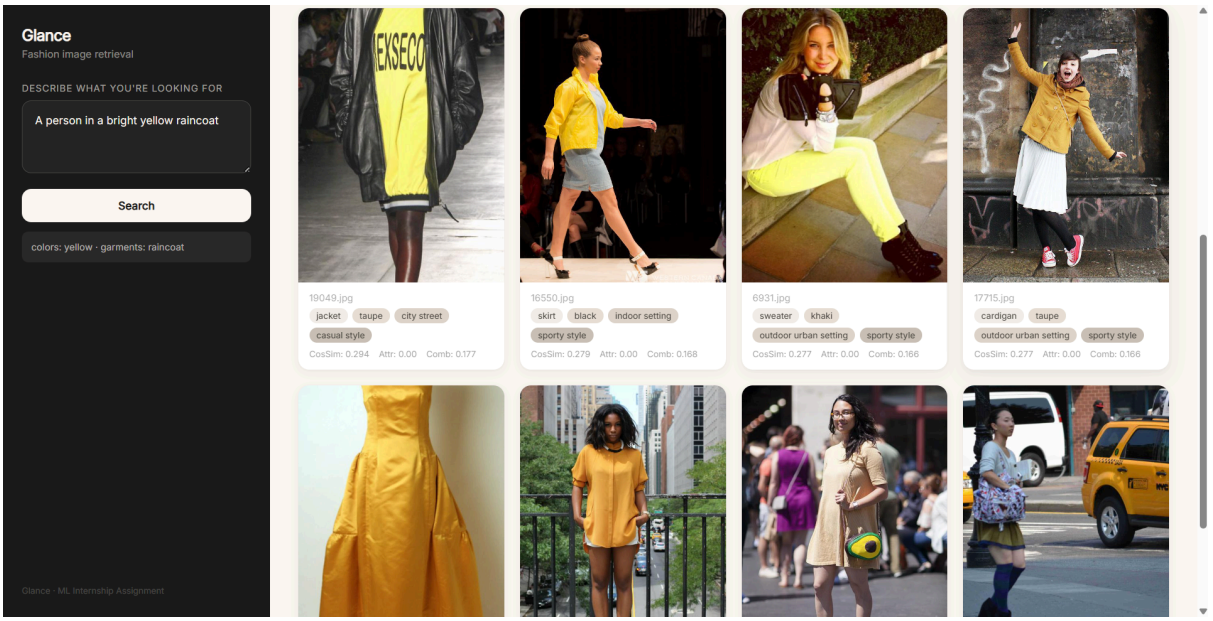
Although the proposed retrieve-and-rerank pipeline improves retrieval performance over a Vanilla CLIP baseline, several challenging queries remain unresolved. These failures are primarily caused by **dataset limitations** and the **image-level nature of the current pipeline**, rather than errors in the retrieval algorithm itself.

Query	Issue	Reason
Q1. Bright Yellow Raincoat	No relevant images retrieved.	The sampled dataset contains very few images matching both the **yellow** color and **coat** category. As a result, the required attribute combination is absent from the indexed metadata.
Q3. Blue Shirt in a Park	Partial retrieval.	Images matching the combined attributes (**blue**, **shirt**, and **park**) are extremely limited. The extracted metadata is also image-level, making it difficult to capture all required attributes accurately.

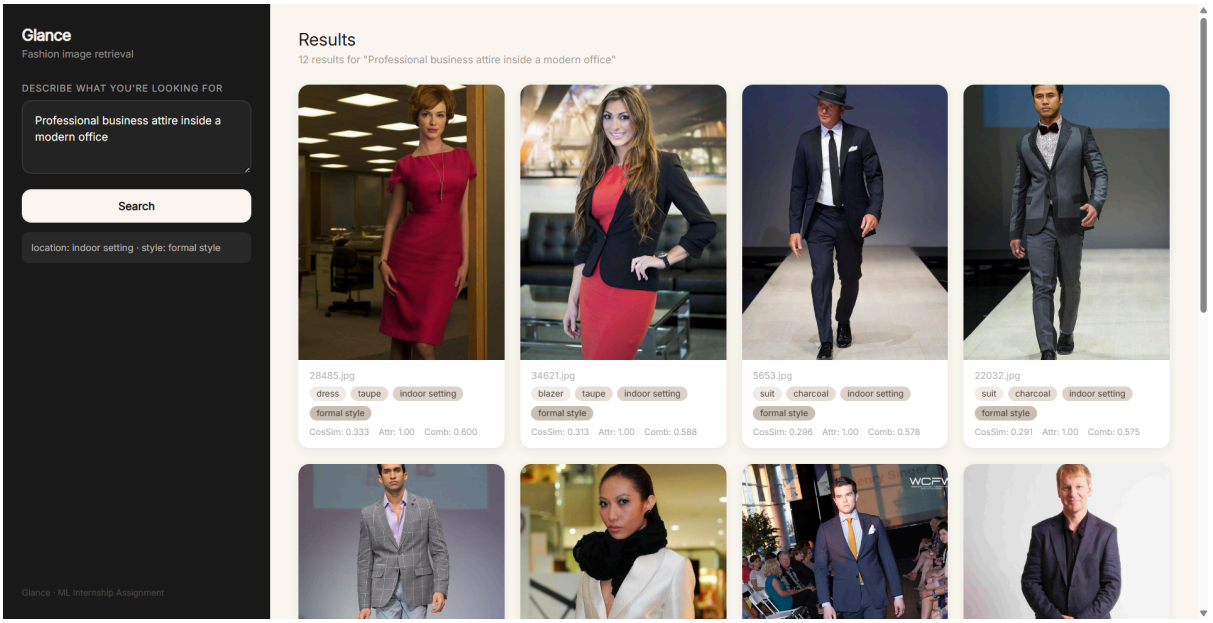
Q5. Red Tie and White Shirt	Partial success (P@5 = 0.60).	The system recognizes the required colors and garments but cannot reliably determine which color belongs to which garment because it uses image-level metadata instead of object-level annotations.

9. Demonstration Results

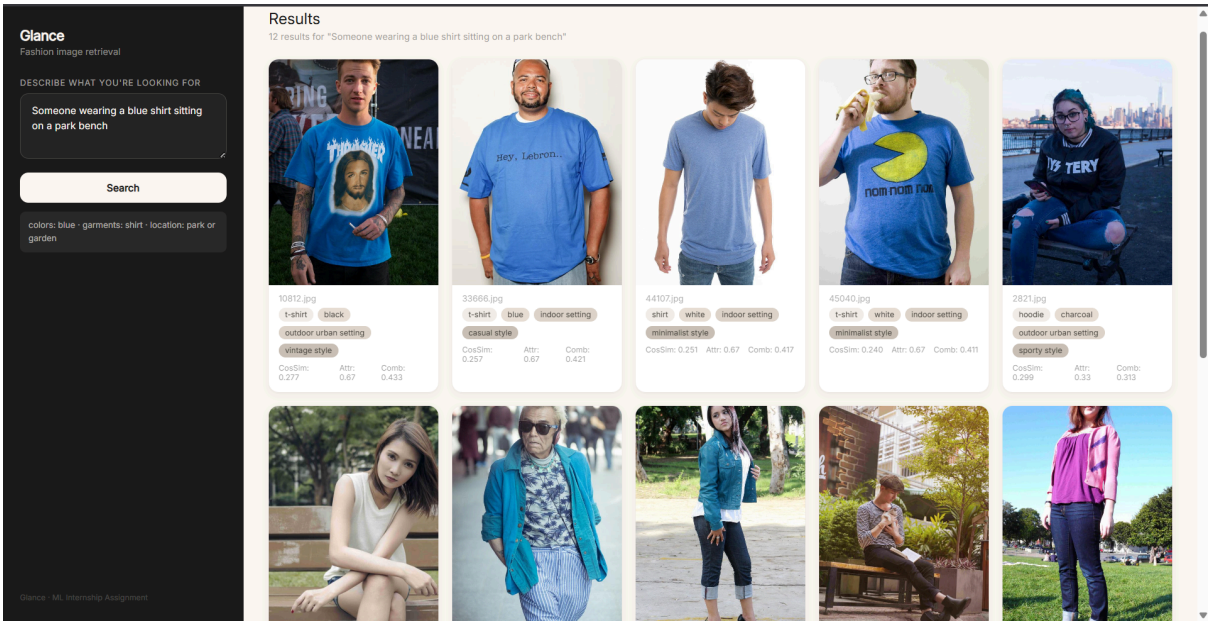
Query 1 -> "A person in a bright yellow raincoat."



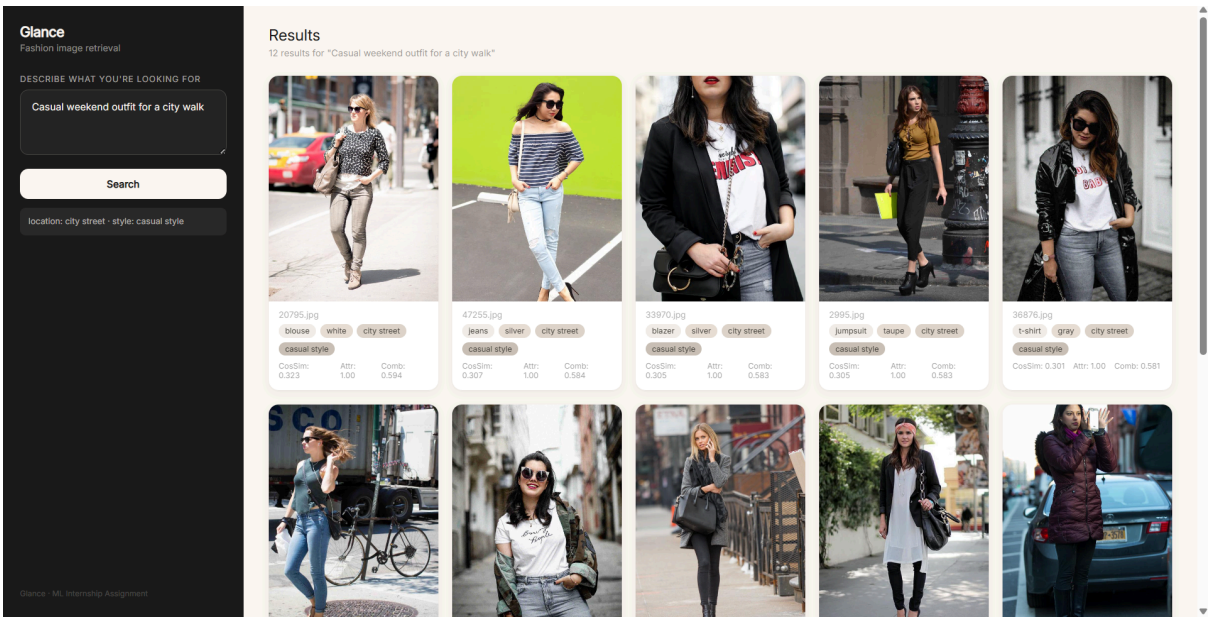
Query 2 ->"Professional business attire inside a modern office."



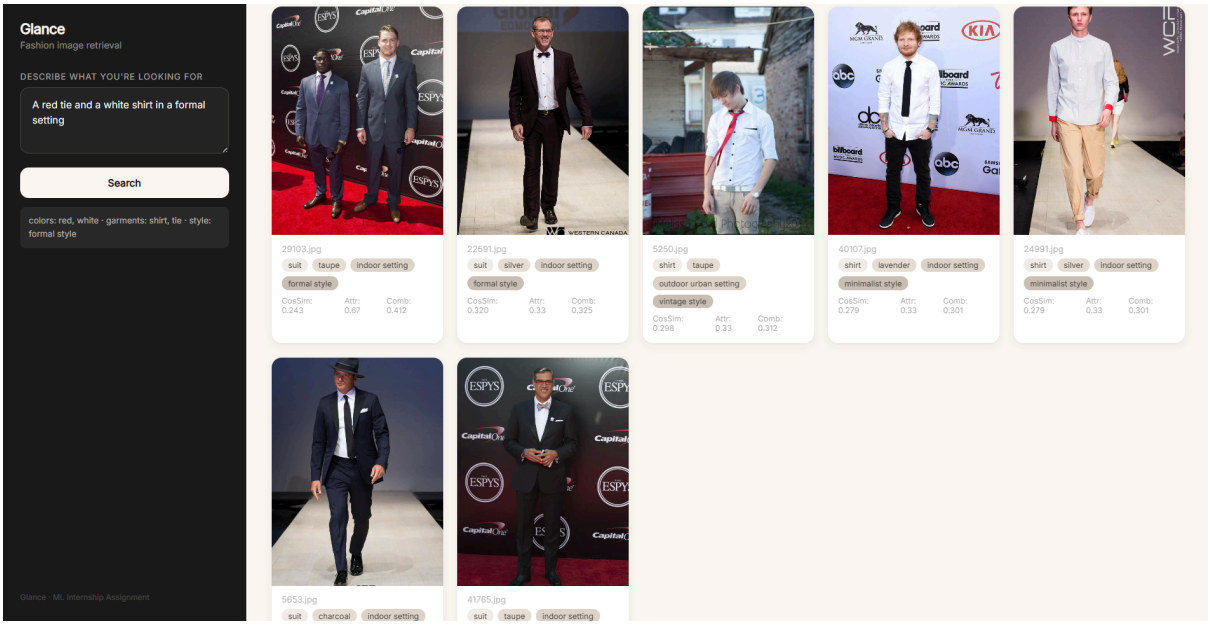
Query 3 ->"Someone wearing a blue shirt sitting on a park bench."



Query 4 ->"Casual weekend outfit for a city walk."



Query 5 ->"A red tie and a white shirt in a formal setting."



10. GitHub Repository

Repository URL ;-

https://github.com/spro047/Glance-Internship-Assignment.git

Repository structure

Glance/

```
├── app/              # Backend & User Interface
├── data/             # Dataset, embeddings, metadata & vector index
├── docs/             # Project documentation and report
├── indexer/          # Image indexing pipeline
├── retriever/        # Search and reranking pipeline
├── evaluation/       # Performance evaluation
├── README.md
└── requirements.txt
```

11. Future Work

Object Detection: Use models like **Grounding DINO** to identify individual clothing items. This will help the system correctly match colors with the right garments.

Better Reranking: Replace the current rule-based reranker with a machine learning model to improve the accuracy of search results.

More Evaluation: Test the system with a larger number of queries and a bigger dataset to better measure its performance.

More Fashion Attributes: Add support for more clothing details such as patterns, sleeve length, fabric type, and accessories to make searches more detailed.

Improve Scalability: Optimize the system so it can efficiently search through much larger image collections, such as millions of fashion images.

12. Conclusion

This project demonstrates a multimodal fashion image retrieval system that combines **FashionCLIP embeddings**, **FAISS vector search**, and

attribute-based reranking to improve search accuracy. The system can understand natural language queries by considering clothing type, color, style, and location instead of relying only on image similarity.

The proposed approach produced relevant results for most of the assignment queries and showed clear improvements over a basic semantic search approach. While the system has some limitations with rare clothing combinations and object-level attribute matching, it provides a scalable and modular foundation that can be further improved with better datasets and more advanced models.

Overall, the project successfully meets the assignment objectives and demonstrates an end-to-end retrieval pipeline that is accurate, efficient, and easy to extend for future improvements.